

MANJEET TIWARI

Java Backend Developer · Spring Boot

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SUMMARY

Final-year B.Tech Computer Science & Technology student at GGSIPU (graduating 2027), targeting Backend Developer, Java Developer, and Software Engineer roles. Skilled in Java, Spring Boot, REST APIs, PostgreSQL, JPA/Hibernate, Spring Security, and JWT. Built SAAKH, a financial transfer backend with a double-entry ledger, idempotency, concurrency control, and a Kafka transactional outbox.

EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi, India 2023 – Expected June 2027
B.Tech, Computer Science & Technology · **GPA 8.9/10 overall** · 9.23/10 (6th Semester)
Class XII (CBSE) 82.8% | *March 2023*
Class X (CBSE) 93.16% | *March 2021*

TECHNICAL SKILLS

- **Languages:** Java, TypeScript, SQL, Python, JavaScript (fundamentals)
- **Backend:** Spring Boot, Spring Security, JPA/Hibernate, REST APIs, JWT, NestJS, TypeORM
- **Data & Messaging:** PostgreSQL, MySQL, H2, Flyway, Apache Kafka
- **Platform & Testing:** Docker, Docker Compose, GitHub Actions, Git, JUnit, MockMvc, Testcontainers
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Concurrency
- **Concepts:** Transactions, Idempotency, Transactional Outbox, Pessimistic Locking, RBAC, Cryptography
- **Familiar:** Redis, MongoDB, FastAPI, Kubernetes, SonarQube, AWS (EC2)

EXPERIENCE

- **AI & Generative AI Intern – IBM SkillsBuild** Aug 2026 – Present
Generative AI, Agentic AI, Machine Learning, IBM watsonx, watsonx Orchestrate
 - Studying machine learning foundations and generative AI: large language models, prompt design, and responsible-use limits.
 - Composing agents from tools and skills in IBM watsonx Orchestrate to automate multi-step workflows rather than one-shot prompts.
- **Backend Developer Intern – PearlThoughts** Jul 2026 – Aug 2026
NestJS, TypeScript, PostgreSQL, TypeORM, REST APIs, JWT
 - Built *Schedula*, a doctor-patient appointment backend: JWT auth, onboarding, recurring availability with date overrides.
 - Implemented *STREAM* and *WAVE* booking modes on PostgreSQL with **pessimistic locking**, a 30-minute cancel/reschedule cutoff, and next-available-slot suggestions.

PROJECTS

- **SAAKH – Financial Transfer & Reconciliation Backend** [Source Code](#)
Java 21, Spring Boot 3.5, PostgreSQL, Flyway, JPA/Hibernate, Kafka, Docker, Testcontainers (47 tests), GitHub Actions
 - Posted each transfer in one PostgreSQL transaction – debit/credit, transfer row, immutable **double-entry ledger**, idempotency record, unpublished outbox event; Kafka never sits on the money path.
 - Locked accounts with ordered `SELECT ... FOR UPDATE`, removing lost updates and $A \rightarrow B$ / $B \rightarrow A$ **deadlocks** under concurrent transfers.
 - Enforced idempotency with an **Idempotency-Key** plus SHA-256 body fingerprint: identical replays return the original transfer, conflicting reuse is rejected with 409.
 - Published events only after commit via a **transactional outbox** (`SKIP LOCKED`, stable event IDs, transferId partition key, retries) at-least-once, with detect-only reconciliation.
- **UPI Offline Mesh Payment System – deferred settlement (simulation)** [Source Code](#)
Java, Spring Boot, RSA-OAEP, AES-256-GCM, SHA-256, H2
 - Simulated offline payments as encrypted packets hopping a software mesh of untrusted devices until an internet-connected bridge uploads them for settlement (not NPCI/UPI infrastructure).
 - Protected packets with hybrid **RSA-OAEP + AES-256-GCM**, GCM tamper detection, signedAt freshness windows, and SHA-256 hashing.
 - Blocked **double settlement** with an atomic hash claim and a unique packet-hash constraint; duplicates and replays are rejected.
- **Common Login Module – JWT Authentication & Role-Based Access Control** [Source Code](#)
Java, Spring Boot, Spring Security, JWT, JPA/Hibernate, BCrypt, H2
 - Built a reusable login module with a custom `JwtAuthenticationFilter`, `CustomUserDetailsService`, `BCrypt` password hashing, and `SecurityContextHolder`-based request authorization.
 - Separated authentication from authorization using **USER / MODERATOR / ADMIN roles**, method-level `@PreAuthorize` (including admin-or-self), and global JSON 401/403/validation handling.

ACHIEVEMENTS & CERTIFICATIONS

- Microsoft TechJam 2.0: **advanced to Round 3** with Team S-Squad, Microsoft Sovereign Office, Noida (Nov 2025); Smart India Hackathon 2025 participant, Team quantum_Coders (Sep 2025).
- Certifications: Spring 5 Basics with Spring Boot and Spring Microservices – Infosys Springboard (Apr 2026); Web Development Professional – MTF Institute (Jun 2026).